



SIMATS ENGINEERING
ESTIMATION EXERCISE QUESTIONS

NAME: T.CHANDANA

REG.NO: 192472059

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1.The delivery times of five online orders are: 20, 25, 30, 35, 40 minutes. Calculate the mean and population variance.

Answer:

Given Data:

20, 25, 30, 35, 40

Number of observations (N) = 5

Mean:

Formula:

$$\text{Mean} = \Sigma X / N$$

$$\text{Mean} = (20 + 25 + 30 + 35 + 40) / 5$$

$$\text{Mean} = 150 / 5$$

$$\text{Mean} = 30 \text{ minutes}$$

Population Variance:

Formula:

$$\text{Population Variance} = \Sigma(X - \mu)^2 / N$$

Here, $\mu = 30$.

X	X - 30	(X - 30) ²
20	-10	100
25	-5	25
30	0	0
35	5	25
40	10	100
Total		250

$$\text{Population Variance} = 250 / 5$$

$$\text{Population Variance} = 50 \text{ minutes}^2$$

$$\text{Mean} = 30 \text{ minutes}$$

$$\text{Population Variance} = 50 \text{ minutes}^2$$

2. A university has 5,000 students. A random sample of 100 students is selected to estimate the average daily study time. Explain why sampling is used instead of surveying all 5,000 students.

Answer:

Given:

Population = 5,000 students

Sample = 100 students

Explanation:

Sampling is the process of selecting a smaller group from a large population to obtain information about the entire population.

In this case, surveying all 5,000 students would require more time, money and effort. Instead, a random sample of 100 students can be selected. If the sample is representative of the university population, the average study time of these 100 students can be used to estimate the average study time of all 5,000 students.

Reasons for using sampling:

1. **Saves time:** Only 100 students need to be surveyed.
2. **Reduces cost:** Fewer resources are required for data collection.
3. **Easy to manage:** Handling 100 responses is easier than handling 5,000 responses.
4. **Faster analysis:** Results can be obtained more quickly.
5. **Practical:** Sampling is suitable when studying the entire population is difficult.

Sampling is used because it provides a faster, cheaper and more practical method of estimating the characteristics of a large population without surveying every individual.

3. From a sample of 200 customers, 150 customers are satisfied with a product. Calculate the sample proportion of satisfied customers and express it as a percentage.

Answer:

Given:

Total number of customers = 200

Number of satisfied customers = 150

Formula:

Sample Proportion = Number of satisfied customers / Total sample size

Sample Proportion = 150 / 200

Sample Proportion = 0.75

To convert the proportion into percentage:

$$\text{Percentage} = 0.75 \times 100$$

$$\text{Percentage} = 75\%$$

$$\text{Sample Proportion} = 0.75$$

$$\text{Percentage of satisfied customers} = 75\%$$

Interpretation:

This means that 75% of the customers in the sample are satisfied with the product.

4. Calculate the covariance between the following variables. Determine whether the relationship is positive or negative.

X	Y
2	5
4	7
6	9
8	11

Answer:

Step 1: Calculate the mean of X

$$\text{Mean of X} = (2 + 4 + 6 + 8) / 4$$

$$\text{Mean of X} = 20 / 4$$

$$\bar{X} = 5$$

Step 2: Calculate the mean of Y

$$\text{Mean of Y} = (5 + 7 + 9 + 11) / 4$$

$$\text{Mean of Y} = 32 / 4$$

$$\bar{Y} = 8$$

Step 3: Calculate covariance

Formula for population covariance:

$$\text{Cov}(X, Y) = \Sigma[(X - \bar{X})(Y - \bar{Y})] / N$$

X	Y	X - 5	Y - 8	(X - 5)(Y - 8)
2	5	-3	-3	9
4	7	-1	-1	1
6	9	1	1	1
8	11	3	3	9
Total				20

Covariance = $20 / 4$

Covariance = 5

Interpretation:

The covariance is positive because the value is greater than zero.

As X increases, Y also increases.

Covariance = 5

Relationship = Positive

5. The following data represents hours of exercise (X) and calories burned (Y). Calculate the covariance and interpret the result.

Exercise Hours (X)	Calories Burned (Y)
1	200
2	300
3	400
4	500
5	600

Answer:

Step 1: Calculate the mean of X

$$\bar{X} = (1 + 2 + 3 + 4 + 5) / 5$$

$$\bar{X} = 15 / 5$$

$$\bar{X} = 3$$

Step 2: Calculate the mean of Y

$$\bar{Y} = (200 + 300 + 400 + 500 + 600) / 5$$

$$\bar{Y} = 2000 / 5$$

$$\bar{Y} = 400$$

Step 3: Calculate covariance

Formula:

$$\text{Cov}(X, Y) = \Sigma[(X - \bar{X})(Y - \bar{Y})] / N$$

X	Y	X - 3	Y - 400	(X - 3)(Y - 400)
1	200	-2	-200	400
2	300	-1	-100	100
3	400	0	0	0
4	500	1	100	100
5	600	2	200	400
Total				1000

$$\text{Covariance} = 1000 / 5$$

$$\text{Covariance} = 200$$

Interpretation:

The covariance is positive.

This means that as exercise hours increase, calories burned also increase.

For example, when exercise increases from 1 hour to 5 hours, calories burned increase from 200 to 600.

$$\text{Covariance} = 200$$

Relationship = Positive

6. Calculate Pearson's correlation coefficient for the following dataset and interpret the strength and direction of the relationship.

Study Hours (X)	Marks (Y)
1	45
2	50
3	55
4	65
5	70

Answer:

Step 1: Calculate the mean of X

$$\bar{X} = (1 + 2 + 3 + 4 + 5) / 5$$

$$\bar{X} = 15 / 5 = 3$$

Step 2: Calculate the mean of Y

$$\bar{Y} = (45 + 50 + 55 + 65 + 70) / 5$$

$$\bar{Y} = 285 / 5 = 57$$

Step 3: Calculate the required values

X	Y	X - 3	Y - 57	(X - 3) ²	(Y - 57) ²	(X - 3)(Y - 57)
1	45	-2	-12	4	144	24
2	50	-1	-7	1	49	7
3	55	0	-2	0	4	0
4	65	1	8	1	64	8
5	70	2	13	4	169	26
Total				10	430	65

Step 4: Pearson's Correlation Formula

$$r = \Sigma[(X - \bar{X})(Y - \bar{Y})] / \sqrt{[\Sigma(X - \bar{X})^2 \times \Sigma(Y - \bar{Y})^2]}$$

Substituting the values:

$$r = 65 / \sqrt{(10 \times 430)}$$

$$r = 65 / \sqrt{4300}$$

$$r = 65 / 65.574$$

$$r \approx 0.991$$

Interpretation:

The value of Pearson's correlation coefficient is approximately +0.991.

Since the value is very close to +1, there is a very strong positive correlation between study hours and marks.

This means that students who spend more hours studying generally tend to obtain higher marks.

Pearson's Correlation Coefficient = 0.991

Strength = Very Strong

Direction = Positive

Interpretation: There is a very strong positive relationship between study hours and marks.